

Report: Object Detection from Scratch

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1. Abstract

This report details the implementation, training, and evaluation of a custom Object Detection system trained entirely from scratch (without pre-trained weights). The objective was to balance detection accuracy with computational efficiency. We implemented a **Faster R-CNN** architecture with a **MobileNetV2** backbone and conducted a rigorous ablation study across differing model capacities (width multipliers) and data augmentation strategies (Basic vs. Mixup). The final results demonstrate a clear trade-off: reducing model width to 0.5x yields a **36% increase in inference speed** (56 FPS) at the cost of accuracy, while advanced Mixup augmentation improves Mean Average Precision (mAP) by **~5-9%** when training from scratch.

2. Methodology

2.1 Dataset Construction

Per the assignment requirements to use a custom dataset of at least 5 classes, we constructed a subset of the **PASCAL VOC 2012 dataset** which was roughly 30% size of the actual dataset.

- **Classes:** Aeroplane, Bicycle, Car, Person, Dog.
- **Preprocessing:** Images were resized and normalized. No external data or pre-trained embeddings were used.

2.2 Model Architecture

We utilized a two stage architecture:

- **Detector Head:** Faster R-CNN (Region-based Convolutional Neural Network) was chosen for its high localization accuracy.
- **Backbone:** **MobileNetV2** was selected over standard ResNet architectures to prioritize inference speed and low parameter count (11.7 million vs 3.4 million).
- **Initialization:** Weights were initialized randomly so that the model is trained from scratch.

2.3 Experimental Design

To investigate the trade-offs between accuracy and speed, we designed a factorial experiment:

1. **Model Width Scaling (`width_mult`):** We varied the channel width of the backbone () to simulate deployment on devices with varying computational power.
2. **Augmentation Strategy:**
 - *None*: Baseline.
 - *Basic*: Random Horizontal Flip, Color Jitter.
 - *Mixup*: A convex combination of two images and their labels to encourage robust feature learning as well as getting smooth decision boundaries.

2.4 Training Configuration

- **Optimizer:** SGD with Momentum (0.9).
- **Learning Rate:** 0.005 (Linear Warmup employed).
- **Epochs:** 50.
- **Batch Size:** 4.

3. Experimental Results

The following table summarizes the performance of all 9 configurations. Metrics reported are **mAP@50** (Mean Average Precision at IoU 0.5 i.e 50% overlap between prediction and ground truth), **mAP (0.5:0.95)** which measures the IoU from 50% to 95% and averages them, **FPS** (Frames Per Second during inference), **Model Size** in Megabytes and **Training Time** in Mins

Model Width	Augmentation	mAP (IoU=0.50)	mAP (0.5:0.95)	Inference FPS	Model Size (MB)	Train Time (mins)
0.5x	Basic	0.5146	0.2196	56.16	308.37	216.3
0.5x	Mixup	0.5783	0.2640	47.03	308.37	216.3
0.5x	None	0.5597	0.2560	47.20	308.37	216.3
0.75x	Basic	0.5461	0.2337	49.74	310.92	238.0
0.75x	Mixup	0.5765	0.2717	44.08	310.92	238.0
0.75x	None	0.5573	0.2554	42.26	310.92	238.0
1.0x	Basic	0.5834	0.2590	47.12	314.23	247.7
1.0x	Mixup	0.6095	0.2949	41.14	314.23	247.7
1.0x	None	0.5556	0.2454	41.47	314.23	247.7

4. Discussion & Analysis

4.1 Accuracy vs. Speed Trade-off

- **Impact of Width Scaling:** Reducing the backbone width from 1.0x to 0.5x resulted in a 36% increase in FPS (41.14 56.16). However, this efficiency gain came with a performance penalty: the mAP dropped from 0.6095 to 0.5146 (Basic Augmentation), representing a ~15% loss in detection accuracy.
- Thus, showing that the 0.5x model is suitable for real-time edge applications where latency is critical, while the 1.0x model is necessary for scenarios requiring precise localization.

4.2 The Efficacy of Mixup for Training from Scratch

One of the most significant findings is the impact of Mixup augmentation.

- **Observation:** In the 1.0x configuration, using Mixup improved mAP from 0.5556 (None) to 0.6095 (Mixup).
- It can be inferred that training from scratch usually suffers from overfitting due to the lack of robust pre-trained features. Mixup effectively acts as a strong regularizer, forcing the model to learn linear interpolations between classes rather than memorizing training samples making the decision boundaries smoother and robust to outliers. This proved superior to "Basic" geometric augmentations.

4.3 Model Size Efficiency

Despite the architectural differences, the model size remained relatively stable (~308MB - 314MB). This indicates that the majority of the parameter count in a Faster R-CNN lies in the detection head (RPN and Classifier), not the MobileNet backbone. Therefore, reducing backbone width is an effective strategy for speed (FLOPs) reduction, but less effective for storage reduction.

5. Conclusion

We successfully built and trained a Faster R-CNN detector from scratch, fulfilling all assignment requirements. Through our ablation study, we identified that the 1.0x Width + Mixup configuration provides the optimal accuracy (60.9% mAP), while the 0.5x Width configuration offers a viable high-speed alternative (56 FPS). The experiments confirm that advanced data augmentation is critical when training deep learning models without pre-trained weights.